

Problem 1)

$$a) A = \begin{bmatrix} 1 & 6 \\ 5 & 4 \\ 9 & 2 \end{bmatrix}; B = \begin{bmatrix} 2 & 8 \\ 4 & 3 \\ 5 & 11 \end{bmatrix} \quad C = \begin{bmatrix} -2 & 2 & 5 \\ -2 & 6 & -4 \end{bmatrix}$$

$$\rightarrow D = A \cdot B = \begin{bmatrix} 3 & 14 \\ 9 & 7 \\ 14 & 13 \end{bmatrix}$$

$$\rightarrow D = AC = \begin{bmatrix} -14 & 38 & -19 \\ -18 & 34 & 9 \\ -22 & 30 & 37 \end{bmatrix}$$

$$\rightarrow D = A \cdot B = \begin{bmatrix} -1 & -2 \\ 1 & 1 \\ 4 & -9 \end{bmatrix}$$

$$\rightarrow D = CA = \begin{bmatrix} 53 & 6 \\ -8 & 4 \end{bmatrix}$$

$$b) A = \begin{bmatrix} \frac{3}{5} & \frac{4}{5} \\ -\frac{4}{5} & \frac{3}{5} \end{bmatrix}, \text{ show that } A^T \cdot A^{-1} \text{ \& } A \text{ is orthogonal} \quad \det(A) = \frac{3}{5} \cdot \frac{3}{5} - \frac{4}{5} \left(-\frac{4}{5}\right)$$

$$A = \begin{bmatrix} a & b \\ -c & d \end{bmatrix}$$

$$A^T = \begin{bmatrix} \frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{bmatrix}$$

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} \frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{bmatrix} = A^{-1}$$

$$= \frac{9}{25} + \frac{16}{25} = 1$$

$$c) \vec{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \alpha = 60^\circ \text{ ccw}$$

$$\vec{b} = R \vec{a} \quad ?$$

$$R = \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} = R$$

$$\vec{b} = \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} - \sqrt{3} \\ \frac{\sqrt{3}}{2} + 1 \end{bmatrix} = \vec{b}$$

$$d) \begin{aligned} 4x_1 + 2x_2 &= 6 \\ 3x_1 - x_2 &= 7 \end{aligned}$$

$$\text{soe: } \begin{bmatrix} 4 & 2 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 6 \\ 7 \end{bmatrix}$$

$$A^{-1} = \frac{1}{-10} \begin{bmatrix} -1 & -2 \\ -3 & 4 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = -\frac{1}{10} \begin{bmatrix} -1 & -2 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} 6 \\ 7 \end{bmatrix} = -\frac{1}{10} \begin{bmatrix} -20 \\ 10 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \vec{x}$$

$$e) \begin{aligned} 4x_1 + 2x_2 &= 6 \\ -2x_1 - x_2 &= 7 \end{aligned}$$

$$\text{soe: } \begin{bmatrix} 4 & 2 \\ -2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 6 \\ 7 \end{bmatrix}$$

$$\text{solvable if } \det(A) \neq 0: \det(A) = -4 - (-4) = 0$$

$\Rightarrow$ not solvable

$$f) \begin{cases} 2x_1 - x_2 = 2 \\ x_1 - 3x_2 + x_3 = -2 \\ -x_1 + x_2 - 3x_3 = -6 \end{cases} \Rightarrow \begin{bmatrix} 2 & -1 & 0 \\ 1 & -3 & 1 \\ -1 & 1 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ -6 \end{bmatrix}$$

$$\begin{array}{ccc|c} 1 & -3 & 1 & -2 \\ -1 & 1 & -3 & -6 \\ 2 & -1 & 0 & 2 \end{array}$$

$$\begin{array}{ccc|c} 1 & -3 & 1 & -2 \\ 2II-II & 0 & 1 & -6 \\ III-2I & 0 & 5 & -2 \end{array}$$

$$\begin{array}{ccc|c} 1 & -3 & 1 & -2 \\ 0 & 1 & -6 & -10 \\ III-5II & 0 & 0 & 56 \end{array} \Rightarrow \begin{aligned} 28x_3 &= 56 \Rightarrow x_3 = 2 \\ x_2 - 6 \cdot 2 &= -10 \Rightarrow x_2 = 2 \\ x_1 - 3 \cdot 2 + 2 &= -2 \Rightarrow x_1 = 2 \end{aligned}$$

in rows:

$\begin{bmatrix} 2 & -1 & 0 \\ 1 & -3 & 1 \\ -1 & 1 & -3 \end{bmatrix} \Rightarrow$ converges

$$g) A = \begin{bmatrix} 2 & -1 & 0 \\ 1 & -3 & 1 \\ -1 & 1 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ -6 \end{bmatrix}$$

$$\begin{aligned} 2x_1 - x_2 &= 2 \Rightarrow x_1 = \frac{2 + x_2}{2} \\ x_1 - 3x_2 + x_3 &= -2 \Rightarrow x_2 = \frac{x_1 + x_3 + 2}{3} \\ -x_1 + x_2 - 3x_3 &= -6 \Rightarrow x_3 = \frac{x_1 - x_2 + 6}{3} \end{aligned}$$

$$k=1: x^{(0)} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x_1^{(1)} = 1$$

$$x_2^{(1)} = \frac{2}{3}$$

$$x_3^{(1)} = 2$$

$$k=2:$$

$$x_1^{(2)} = \frac{2 + \frac{2}{3}}{2} = \frac{4}{3}$$

$$x_2^{(2)} = \frac{1 + 2 + \frac{4}{3}}{3} = \frac{5}{3}$$

$$x_3^{(2)} = \frac{\frac{4}{3} - \frac{5}{3} + 6}{3} = \frac{17}{9}$$

$$k=3:$$

$$x_1^{(3)} = \frac{2 + \frac{5}{3}}{2} = \frac{11}{6}$$

$$x_2^{(3)} = \frac{\frac{11}{6} + \frac{17}{9} + 2}{3} = \frac{47}{27}$$

$$x_3^{(3)} = \frac{\frac{11}{6} - \frac{47}{27} + 6}{3} = \frac{19}{9}$$

$$k=4:$$

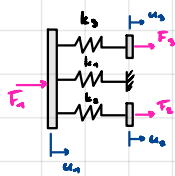
$$x_1^{(4)} = \frac{2 + \frac{47}{27}}{2} = \frac{101}{54} \approx 1.87$$

$$x_2^{(4)} = \frac{\frac{11}{6} + \frac{47}{27} + 2}{3} = \frac{103}{54} \approx 1.98$$

$$x_3^{(4)} = \frac{\frac{47}{27} - \frac{101}{54} + 6}{3} = \frac{319}{162} \approx 1.97$$

$\Rightarrow$ It does converge to the exact solution $\begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$

Problem 2)



a)

$$V = \frac{1}{2} k_1 u_1^2 + \frac{1}{2} k_2 (u_1 - u_2)^2 + \frac{1}{2} k_3 (u_1 - u_3)^2$$

$$\frac{\partial V}{\partial u_1} = k_1 u_1 + k_2 (u_1 - u_2) + k_3 (u_1 - u_3)$$

$$\frac{\partial V}{\partial u_2} = -k_2 (u_1 - u_2)$$

$$\frac{\partial V}{\partial u_3} = -k_3 (u_1 - u_3)$$

$$\text{SDE: } \underline{\tilde{K}} \underline{\tilde{u}} = \underline{\tilde{F}} : \underbrace{\begin{bmatrix} k_1 + k_2 + k_3 & -k_2 & -k_3 \\ -k_2 & k_2 & 0 \\ -k_3 & 0 & k_3 \end{bmatrix}}_{\underline{\tilde{K}}} \underbrace{\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}}_{\underline{\tilde{u}}} = \underbrace{\begin{bmatrix} F_1 \\ F_2 \\ F_3 \end{bmatrix}}_{\underline{\tilde{F}}}$$

b) if $\det(\underline{\tilde{K}}) \neq 0 \Rightarrow$ unique solution

$$\begin{aligned} \det(\underline{\tilde{K}}) &= k_3 ((k_1 + k_2 + k_3) \cdot k_2 - k_2^2) - k_3 k_2 k_3 \\ &= k_3 (k_1 k_2 + k_2^2 + k_2 k_3 - k_2^2) - k_3^2 k_2 \\ &= k_1 k_2 k_3 + k_3^2 k_2 - k_3^2 k_2 = k_1 k_2 k_3 = \det(\underline{\tilde{K}}) \end{aligned}$$

⇓
System has a unique solution if none of the three stiffnesses are zero

c) $k_i = 10, i=1,2,3$

assume $u_3 = 0, F_1 = 1,5, F_2 = 0,5$

$$\begin{aligned} &\begin{bmatrix} k_1 + k_2 + k_3 & -k_2 & -k_3 \\ -k_2 & k_2 & 0 \\ -k_3 & 0 & k_3 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} F_1 \\ F_2 \\ F_3 \end{bmatrix} \\ \Rightarrow &\begin{bmatrix} 30 & -10 & -10 \\ -10 & 10 & 0 \\ -10 & 0 & 10 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1,5 \\ 0,5 \\ F_3 \end{bmatrix} \quad \begin{aligned} \text{I} + \text{II} : 20u_1 &= 2 \Rightarrow u_1 = \frac{1}{10} = 0,1 \\ \Rightarrow -10 \cdot \frac{1}{10} + 10u_2 &= 0,5 \Rightarrow u_2 = \frac{1,5}{10} = 0,15 \\ \Rightarrow -10 \cdot \frac{1}{10} &= F_3 = -1 \end{aligned} \end{aligned}$$